

Displays

D1:

$$\mathcal{Z}(\mathbf{c}, \mathbf{G}, \mathbf{A}, \mathbf{b}) = \{ \mathbf{c} + \mathbf{G}\mathbf{z} \mid \mathbf{A}\mathbf{z} = \mathbf{b}, \|\mathbf{z}\|_\infty \leq 1 \}. \quad (1)$$

D2:

$$\mathbf{x}_{k+1} = \mathbf{f}(\mathbf{x}_k, \mathbf{u}_k) + \mathbf{w}_k. \quad (2)$$

D3:

$$R_k = \{ \mathbf{x}_k \in \mathbb{R}^n \mid \mathbf{x}_{j+1} = \mathbf{f}(\mathbf{x}_j, \mathbf{u}_j) + \mathbf{w}_j, \mathbf{x}_0 \in X_0, \\ \mathbf{u}_j \in U_j, \text{ and } \mathbf{w}_j \in W, \forall j = 0, \dots, k-1 \}. \quad (3)$$

D4:

$$\mathbf{X}_- = \left[\mathbf{x}_0^{(1)}, \dots, \mathbf{x}_{t_1-1}^{(1)}, \mathbf{x}_0^{(2)}, \dots, \mathbf{x}_0^{(q)}, \dots, \mathbf{x}_{t_q-1}^{(q)} \right], \quad (4a)$$

D5:

$$\mathbf{X}_+ = \left[\mathbf{x}_1^{(1)}, \dots, \mathbf{x}_{t_1}^{(1)}, \mathbf{x}_1^{(2)}, \dots, \mathbf{x}_1^{(q)}, \dots, \mathbf{x}_{t_q}^{(q)} \right], \quad (4b)$$

D6:

$$\mathbf{U}_- = \left[\mathbf{u}_0^{(1)}, \dots, \mathbf{u}_{t_1-1}^{(1)}, \mathbf{u}_0^{(2)}, \dots, \mathbf{u}_0^{(q)}, \dots, \mathbf{u}_{t_q-1}^{(q)} \right]. \quad (4c)$$

D7:

$$Z_\cap = \mathcal{Z} \left(\mathbf{c}_1, [\mathbf{G}_1, \mathbf{0}], \begin{bmatrix} \mathbf{A}_1 & \mathbf{0} \\ \mathbf{0} & \mathbf{A}_2 \\ \mathbf{G}_1 & -\mathbf{G}_2 \end{bmatrix}, \begin{bmatrix} \mathbf{b}_1 \\ \mathbf{b}_2 \\ \mathbf{c}_2 - \mathbf{c}_1 \end{bmatrix} \right). \quad (5)$$

D8:

$$v^\star = \min_{\mathbf{z}, v} \{ v \mid \mathbf{A}_\cap \mathbf{z} = \mathbf{b}_\cap \text{ and } |\mathbf{z}| \leq v \}, \quad (6)$$

D9:

$$\nabla_{\mathbf{u}_h} v^\star = \nabla_{\hat{\mathbf{c}}_k} v^\star \nabla_{\hat{\mathbf{c}}_{k-1}} \hat{\mathbf{c}}_k \left(\prod_{j=h+2}^{j=k-1} \nabla_{\hat{\mathbf{c}}_{j-1}} \hat{\mathbf{c}}_j \right) \nabla_{\mathbf{u}_h} \hat{\mathbf{c}}_{h+1}, \quad (7)$$

D10:

$$\nabla_{\hat{\mathbf{c}}_{k-1}} \hat{\mathbf{c}}_k = (\mathbf{M}_{k-1})_{(1:1+n), (1:1+n)}, \text{ and} \quad (8a)$$

D11:

$$\nabla_{u_{k-1}} \hat{\mathbf{c}}_k = (\mathbf{M}_{k-1})_{:, (n+1:n+1+m)}, \quad (8b)$$

Inline

I1: ¹

I2: ²

I3: ³

I4: ⁴

I5: ϵ

I6: $\ast$

I7: n

I8: $\mathbb{R}^n$

I9: $\mathbb{N}$

I10: m

I11: $n:m$

I12: i

I13: j

I14: $\mathbf{A}$

I15: $(\mathbf{A})_{i,j}$

I16: $(\mathbf{A})_{:,j}$

I17: $\mathbf{a}$

I18: $(\mathbf{a})_i$

I19: $n \times m$

I20: $\mathbf{1}_{n \times m}$
 I21: $\mathbf{A} \in \mathbb{R}^{n \times m}$
 I22: $\mathbf{x} \in \mathbb{R}^n$
 I23: $\mathbf{A} - \mathbf{x} = \mathbf{A} - \mathbf{x} \mathbf{1}_{1 \times m}$
 I24: $\text{diag}(\cdot)$
 I25: A
 I26: B
 I27: $A + B = \{\mathbf{a} + \mathbf{b} \mid \mathbf{a} \in A, \mathbf{b} \in B\}$
 I28: $A \times B = \{(\mathbf{a}, \mathbf{b}) \mid \mathbf{a} \in A, \mathbf{b} \in B\}$
 I29: $\mathbf{c} \in \mathbb{R}^n$
 I30: $\mathbf{G} \in \mathbb{R}^{n \times n_g}$
 I31: $\mathbf{A} \in \mathbb{R}^{n_c \times n_g}$
 I32: $\mathbf{b} \in \mathbb{R}^{n_c}$
 I33: $\|\mathbf{z}\|_\infty \leq 1$
 I34: $\mathcal{Z}(\mathbf{c}, \mathbf{G})$
 I35: $Z = \mathcal{Z}(\mathbf{c}, \mathbf{G}) \subset \mathbb{R}^n$
 I36: L
 I37: $LZ = \mathcal{Z}(L\mathbf{c}, L\mathbf{G})$
 I38: $-Z = -1Z$
 I39: $Z_1 = \mathcal{Z}(\mathbf{c}_1, \mathbf{G}_1)$
 I40: $Z_2 = \mathcal{Z}(\mathbf{c}_2, \mathbf{G}_2)$
 I41: $Z_1 + Z_2 = \mathcal{Z}(\mathbf{c}_1 + \mathbf{c}_2, [\mathbf{G}_1, \mathbf{G}_2])$
 I42: $\underline{\mathbf{1}} \in \mathbb{R}^n$
 I43: $\bar{\mathbf{1}}$
 I44: $Z = \mathcal{Z}(\underline{\mathbf{1}}, \bar{\mathbf{1}}) \subset \mathbb{R}^n$
 I45: $\frac{1}{2}(\underline{\mathbf{1}} + \bar{\mathbf{1}})$
 I46: $\text{diag}(\frac{1}{2}(\bar{\mathbf{1}} - \underline{\mathbf{1}}))$
 I47: $\mathbf{x}_k \in X \subset \mathbb{R}^n$
 I48: $k \in \mathbb{N}$
 I49: X
 I50: $\mathbf{u}_k$
 I51: $U_k \subseteq U$
 I52: k
 I53: $U \subset \mathbb{R}^m$
 I54: $\mathbf{w}_k \in W \subset \mathbb{R}^n$
 I55: W
 I56: $\mathbf{f} : X \times U \times W \rightarrow X$
 I57: $\mathbf{f}$
 I58: $L^\star$
 I59: $\forall$
 I60: $\mathbf{z}_1, \mathbf{z}_2 \in \mathbb{R}^{n+m}$
 I61: $\mathbf{z}_j = (\mathbf{x}_j, \mathbf{u}_j)$
 I62: $\|\mathbf{f}(\mathbf{z}_1) - \mathbf{f}(\mathbf{z}_2)\| \leq L^\star \|\mathbf{x}_1 - \mathbf{x}_2\|$
 I63: $\mathbf{x}_0$
 I64: $X_0 \subset \mathbb{R}^n$
 I65: $n_{\text{brk}} \in \mathbb{N}$
 I66: $\mathbf{u}_{\text{brk}} \in U$
 I67: $\mathbf{x}_k$
 I68: $\mathbf{x}_{k+1} = \mathbf{f}(\mathbf{x}_k, \mathbf{u}_{\text{brk}})$
 I69: $\mathbf{x}_{k+1} = \mathbf{x}_k$
 I70: $\mathbf{w}_k$
 I71: $W = \mathcal{Z}(\mathbf{c}_w, \mathbf{G}_w)$
 I72: $n_{g, w}$
 I73: $X_{\text{obs}} \subset X$
 I74: X_{obs}
 I75: R_k

I76: $\mathbf{u}_j \in U_j \subset \mathbb{R}^m$
 I77: $\mathbf{w}_j \in W$
 I78: $\forall j \in \{0, \dots, k-1\}$
 I79: $X_0 \in \mathbb{R}^n$
 I80: R_k
 I81: $\hat{\mathbf{x}}_k \in \mathbb{R}^{n_{\text{RL}}}$
 I82: $X_{\text{obs}} \subset \mathbb{R}^n$
 I83: $\rho : (\hat{\mathbf{x}}_k, \mathbf{u}_k) \mapsto r_k \in \mathbb{R}$
 I84: ρ
 I85: $\mathbf{p}_k = (\mathbf{u}_j)_{j=k}^{n_{\text{plan}}}$
 I86: $n_{\text{plan}} \in \mathbb{N}$
 I87: $\pi_\theta : \hat{\mathbf{x}}_k \mapsto \mathbf{u}_k$
 I88: θ
 I89: $R_j \cap X_{\text{obs}} = \emptyset$
 I90: $j \geq k$
 I91: $\mathbf{p}_k$
 I92: k^{th}
 I93: $\mathbf{p}_{k-1}$
 I94: π_θ
 I95: μ_ϕ
 I96: n_{iter}
 I97: $\mathbf{p}_0$
 I98: $\hat{\mathbf{x}}_1 \leftarrow$
 I99: $k = 1 : n_{\text{iter}}$
 I100: $\mathbf{p}_k \leftarrow$
 I101: $\hat{R}_k \leftarrow \mathcal{Z}(\mathbf{x}_k, \mathbf{0})$
 I102: $\left(\hat{R}_j\right)_{j=k}^{k+n_{\text{plan}}} \leftarrow \text{reach}\left(\hat{R}_k, \mathbf{p}_k\right)$
 I103: $\hat{R}_j \cap X_{\text{obs}} \neq \emptyset$
 I104: $\mathbf{p}_k \leftarrow \text{adjust}(\mathbf{p}_k, X_{\text{obs}})$
 I105: $\mathbf{u}_k \leftarrow$
 I106: $r_k \leftarrow \rho(\hat{\mathbf{x}}_k, \mathbf{u}_k)$
 I107: $\hat{\mathbf{x}}_{k+1} \leftarrow$
 I108: $(\hat{\mathbf{x}}_k, \mathbf{u}_k, r_k, \hat{\mathbf{x}}_{k+1})$
 I109: $\hat{R}_j \supseteq R_j$
 I110: $\hat{R}_0$
 I111: $(\mathbf{u}_j)_{j=k}^{k+n_{\text{plan}}}$
 I112: $(\mathbf{X}_-, \mathbf{X}_+, \mathbf{U}_-)$
 I113: δ
 I114: $Z_\epsilon \leftarrow \mathcal{Z}(\mathbf{0}, \text{diag}((\mathbf{L}^*)_1(\delta)_1/2, \dots, (\mathbf{L}^*)_n(\delta)_n/2))$
 I115: $j = k : (k + n_{\text{plan}})$
 I116: $\mathbf{M}_j \leftarrow (\mathbf{X}_+ - \mathbf{c}_w) \begin{bmatrix} \mathbf{1}_{1 \times t_{\text{total}}} \\ \mathbf{X}_- - \mathbf{x}_j^* \\ \mathbf{U}_- - \mathbf{u}_j \end{bmatrix}^\dagger$
 I117: $\underline{\mathbf{l}} \leftarrow \min_j \left((\mathbf{X}_+)_{:,j} - \mathbf{M}_j \begin{bmatrix} 1 \\ (\mathbf{X}_-)_{:,j} - \mathbf{x}_j^* \\ (\mathbf{U}_-)_{:,j} - \mathbf{u}_j \end{bmatrix} \right)$
 I118: $\bar{\mathbf{l}} \leftarrow$
 I119: $\underline{\mathbf{l}}$
 I120: $Z_L \leftarrow \mathcal{Z}(\underline{\mathbf{l}}, \bar{\mathbf{l}}) - W$
 I121: $U_j \leftarrow \mathcal{Z}(\mathbf{u}_j, \mathbf{0})$
 I122: $\hat{R}_{j+1} \leftarrow \mathbf{M}_j(\mathbf{1} \times (\hat{R}_j - \mathbf{x}_j^*) \times (U_j - \mathbf{u}_j)) + W + Z_L + Z_\epsilon$
 I123: $(\hat{R}_j)_{j=k}^{k+n_{\text{plan}}}$
 I124: q
 I125: $t_i \in \mathbb{N}$

I126: $i = 1, \dots, q$
 I127: $t_{\text{total}} = \sum_i^q t_i$
 I128: $(\mathbf{x}_k^{(i)})_{k=0}^{t_i}$
 I129: $(\mathbf{u}_k^{(i)})_{k=0}^{t_i-1}$
 I130: $i = 1, \dots, q$
 I131: $\mathbf{X}_-$
 I132: $\mathbf{X}_+$
 I133: $\mathbf{U}_-$
 I134: $\mathbf{z}_1 \in X \times U$
 I135: $\mathbf{z}_2 \in X \times U$
 I136: $\|\mathbf{z}_1 - \mathbf{z}_2\|_2 \leq \delta$
 I137: $(L^*)_i$
 I138: $(\delta)_i$
 I139: Z_ϵ
 I140: $\mathbf{p}$
 I141: $\hat{R}_k$
 I142: γ
 I143: t_{max}
 I144: n_{brk}
 I145: $\mathbf{u}_{\text{brk}}$
 I146: $j > k + n_{\text{plan}}$
 I147: $\mathbf{p}_{\text{safe}} \leftarrow \mathbf{p}_k$
 I148: $j = k : (k + n_{\text{plan}} + n_{\text{brk}})$
 I149: $(\hat{R}_j)_{j=k}^{k+n_{\text{plan}}} \leftarrow \text{reach}(\hat{R}_j, \mathbf{p}_{\text{safe}})$
 I150: $v^* \leftarrow$
 I151: $\hat{R}_j \cap X_{\text{obs}}$
 I152: $v^* \leq 1$
 I153: $\mathbf{u}_j \leftarrow \text{proj}_{U_j}(\mathbf{u}_j + \gamma \nabla_{\mathbf{u}_j} v^*)$
 I154: $\hat{R}_j \cap X_{\text{obs}} = \emptyset$
 I155: $\mathbf{x}_n$
 I156: $\mathbf{p}_{\text{safe}} = (\mathbf{u}_j)_{j=k}^{n_{\text{plan}}}$
 I157: $\mathbf{u}_0$
 I158: $Z_1 = \mathcal{Z}(\mathbf{c}_1, \mathbf{G}_1, \mathbf{A}_1, \mathbf{b}_1)$
 I159: $Z_2 = \mathcal{Z}(\mathbf{c}_2, \mathbf{G}_2, \mathbf{A}_2, \mathbf{b}_2)$
 I160: $Z_\cap = Z_1 \cap Z_2 = \mathcal{Z}(\mathbf{c}_\cap, \mathbf{G}_\cap, \mathbf{A}_\cap, \mathbf{b}_\cap)$
 I161: $Z_1 \cap Z_2$
 I162: $|\mathbf{z}|$
 I163: $Z_\cap$
 I164: $v \leq 1$
 I165: Z_1
 I166: Z_2
 I167: $\hat{\mathbf{c}}_k$
 I168: $\mathbf{u}_0, \dots, \mathbf{u}_{k-1}$
 I169: $(\mathbf{z}^*, v^*)$
 I170: $v^* > 1$
 I171: $\nabla_{\mathbf{u}_k} v^*$
 I172: $i = 0, \dots, n_{\text{plan}}$
 I173: $h = k - i$
 I174: $\mathbf{M}_{k-1}$
 I175: $\nabla_{\mathbf{u}_k} v^*$
 I176: $\mathbf{u}_k$
 I177: $\text{proj}_{U_k}(\mathbf{u}_k) = \arg \min_{\mathbf{v} \in U_k} \{\|\mathbf{u}_k - \mathbf{v}\|_2^2\}$
 I178: $k = 0$
 I179: $k > 0$
 I180: $k \geq 0$

I181: 0
 I182: n_{plan}
 I183: $j \geq k + n_{\text{plan}}$
 I184: $\square$
 I185: $X_{\text{goal}} \subset X$
 I186: $[0.00, 0.25]$
 I187: $[-0.5, 0.5]$
 I188: U_k
 I189: 180°
 I190: $n_{\text{brk}} = 6$
 I191: $n_{\text{plan}} = 8$
 I192: 50°
 I193: 360°
 I194: $n_{\text{brk}} = 10$
 I195: $n_{\text{plan}} = 11$
 I196: r
 I197: $\mathbf{x}_k = (x_k, y_k, \dot{x}_k, \dot{y}_k)$
 I198: $X_{\text{safe}} = \{\mathbf{x}_k \in X : |x_k| \leq x_{\text{max}}, |y_k| \leq y_{\text{max}}\}$
 I199: $\|(x_{\text{max}}, y_{\text{max}})\|_2 < r$
 I200: $\rho(\hat{\mathbf{x}}_k, \mathbf{u}_k) = \frac{(\dot{x}_k, \dot{y}_k) \cdot (-y_k, x_k)}{1 + \|(x_k, y_k)\|_2 - r}$
 I201: $\mathbf{M}_j$